

A span-sharing joint extraction framework for harvesting aspect sentiment triplets

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ARTICLE INFO

Article history:

Received 19 September 2021

Received in revised form 13 January 2022

Accepted 3 February 2022

Available online 10 February 2022

Keywords:

Sentiment analysis

Aspect sentiment triplet extraction

Span-sharing

Syntactic dependency

ABSTRACT

Aspect sentiment triplet extraction (ASTE) is the most recent subtask of aspect-based sentiment analysis (ABSA), and it aims to provide a complete solution for ABSA by extracting the triplets of aspect terms, opinion terms and their associated sentiments. However, previous pipeline approaches to ASTE suffer from error propagation in different subtasks, and previous tagging-based joint extraction approaches fail to deal with one-to-many and many-to-one relationship between aspect terms and opinion words in a sentence. To address this, we propose a span-sharing joint extraction (SSJE) framework to extract aspect sentiment triplets from sentences in an end-to-end fashion. In this framework, an aspect term and the corresponding opinion term(s), combined with the sentiment on the aspect, are identified simultaneously in the last step to avoid the error propagation. At the same time, all possible candidate spans are enumerated to be shared by aspect terms and opinion terms to deal with the problem of multiple-word entities and the complex relationship between aspect terms and opinion terms. To verify SSJE's effectiveness, we conduct extensive experiments on two benchmark datasets. The experimental results demonstrate that the proposed SSJE achieves state-of-the-art performances not only for ASTE tasks but also for extracting aspect-opinion pairs.

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1. Introduction

Sentiment analysis, also known as opinion mining, is an important task in natural language processing that aims to analyze sentiments, opinions, evaluations, attitudes and emotions from human language [1–3]. As a fine-grained sentiment analysis task, aspect-based sentiment analysis (ABSA) focuses on answering the 3 “what” questions of a review sentence: “what” are the user’s targeted aspects; “what” words does the user use to express sentiment; and “what” kind of sentiment does the user hold? Specifically, ABSA aims to identify three core opinion entities in reviews: aspect terms, opinion terms and the sentiment associated with each aspect. Aspect terms, sometimes called aspects or opinion targets, are the words or phrases in a sentence representing features or entities toward which users express opinion. Opinion terms, also called opinion words, refer to the terms used to express the user’s opinion explicitly. The sentiment indicates the type of sentiment users express on the aspect, such as positive, neutral and negative. As shown in the example sentence in Fig. 1, *food* and *service* are two aspects, *great* and *dreadful* are two opinion terms and the user has a positive sentiment regarding the *food* and a negative sentiment regarding the *service*. Since ABSA

can provide a complete and detailed view of sentiment regarding different aspects, it has drawn increasing attention in recent years not only in academia but also in industry.

Most existing methods for ABSA focus on individual opinion entities or a combination of two opinion entities; these include, but are not limited to, aspect term extraction (ATE) [4–6], opinion term extraction (OTE) [7–9], aspect-opinion pair extraction (AOPE) [10–12], and aspect sentiment classification (ASC) [13–15]. Fig. 1 shows the differences between the above ABSA subtasks for an example sentence. ATE and OTE extract aspect terms and opinion terms, respectively, and some studies have coextracted these two objects in recent years [16–19]. However, coextraction methods neglect to recognize the relations between aspect terms and opinion terms. AOPE aims at extracting aspect terms and opinion terms in pairs to retain such relations. For example, (*food*, *great*) and (*service*, *dreadful*) are the opinion pairs extracted by AOPE in the above example. Pairwise extraction methods cannot identify aspects, opinion terms and sentiment to obtain a complete solution. At the same time, the context information of opinion pairs is missed, which may reduce the effect of predicting aspect-level sentiment. ASC identifies the aspect together with the user’s sentiment toward the aspect in a sentence, which ignores the words used to express the user’s opinion. Thus, none of the above solvers can provide a complete solution to answer the 3 “what” questions of ABSA.

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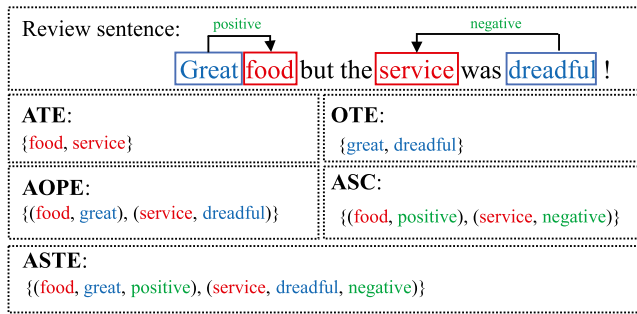


Fig. 1. An example of the differences between ABSA subtasks.

To address this problem, a new task for ABSA, namely, aspect sentiment triplet extraction (ASTE), was recently proposed in [20]; it aims at extracting not only the aspects and the sentiment but also the corresponding opinion terms expressing the sentiment for each aspect, such as the triplets (*food*, *great*, *positive*) and (*service*, *dreadful*, *negative*) in Fig. 1. Peng et al. designed a two-stage framework to extract sentiment triplets [20], in which many aspects with sentiment polarities and various opinion terms are identified separately in the first step. The triplets are obtained based on pairing aspects and opinion terms in the second step. Other pipeline methods for ASTE can be found in [21,22], etc. These pipeline methods break triplet extraction into multiple subtasks, and each subtask identifies one opinion entity within a combination of two opinion entities. Then, the triplets are obtained based on the results generated in the previous steps. Such methods break the interaction within the triplet structure and lead to error propagation in different subtasks. Another drawback is that most existing pipeline methods rely heavily on word-to-word interactions to identify opinion entities, but it is common for aspect and opinion terms to contain multiple words in real-world scenarios. On the other hand, the joint extraction method mainly adopts a sequence tagging mechanism to extract sentiment triplets [23,24], which labels each word of a sentence with a semantic tag to specify the word's role. However, this method becomes difficult to apply when there is a one-to-many or many-to-one relationship between aspect terms and opinion terms in a sentence.

To address the shortcomings of the existing methods of ASTE, we propose an end-to-end span-sharing joint extraction (SSJE) framework to harvest aspect sentiment triplets from review sentences. SSJE first learns word-level representations using a base encoder, and the words' dependency information obtained by a graph convolutional network (GCN) on the syntactic dependency tree of the sentence is integrated into the span representations. Then, all possible spans on the sentence are enumerated and filtered to generate the candidate spans, which can be shared as aspect terms and opinion terms. Finally, two candidate spans, which respectively act as the aspect term and the opinion term, and the corresponding local context are fed into a classifier to generate sentiment triplets in one shot.

Unlike the pipeline method, which extracts different opinion entities or entity pairs at different stages, SSJE identifies the aspect term, opinion term and sentiment on the aspect simultaneously in the last step. In this way, the problem of error propagation encountered by pipeline methods can be effectively avoided. Additionally, unlike the sequence tagging method, we enumerate all possible candidate pairs based on the generated spans, which effectively deals with the one-to-many and many-to-one relations between aspect terms and opinion terms. To validate the effectiveness of our method, we conduct a series

of experiments on multiple ABSA benchmark datasets. The experimental results show that SSJE can efficiently deal with the complex relations between aspect terms and opinion terms in a sentence and outperform the state-of-the-art pipeline methods and joint methods of extracting sentiment triplets for the ASTE task. Moreover, we demonstrate that SSJE achieves remarkable effectiveness in extracting aspect-opinion pairs for the AOE task.

The main contributions of our work can be summarized as follows:

- We propose a SSJE framework to harvest aspect sentiment triplets from review sentences in an end-to-end fashion, which can effectively avoid the error propagation problem in different subtasks and deal with multiple-word entities and the complex relations of one-to-many and many-to-one between aspect terms and opinion terms.
- Considering the importance of the dependency relationship between spans, SSJE introduces the syntactic dependency information obtained by graph convolution into our model to reduce the influence between different sentiment triplets.
- We conduct extensive experiments on multiple public ABSA datasets to show that the proposed SSJE achieves state-of-the-art performances, not only for ASTE tasks but also for AOE tasks.

The remainder of this paper is organized as follows: The related work is summarized in Section 2. We formalize the ASTE task and present the proposed model in detail in Section 3. A series of experiments are conducted to evaluate the proposed model in Section 4. Finally, Section 5 concludes our work.

2. Related work

Compared with conventional sentiment analysis, which predicts the overall sentiment type of a given text, ABSA provides more granular sentiment analysis. The ASTE task represents the latest research in the field of ABSA, and is aimed at extracting sentiment triplets from core objects in sentiment analysis. It was proposed by Peng et al. [20]. The existing methods of ASTE can be roughly divided into two categories: pipeline methods and joint extraction methods.

- **Pipeline extraction methods.** There are multiple subtasks in ABSA, but ASTE and AOE are the most relevant ones since aspect-based sentiment type can be detected from aspect terms and opinion terms in the AOE task to extract sentiment triplets. Therefore, it is also the most direct pipeline method to extract sentiment triplets. For example, RINANTE [18], CMLA [16] and Li-unified-R [25] were extended in [20] to achieve the extraction of sentiment triplets and were compared with the two-stage sentiment triplet extraction method proposed in [20]. For the AOE task, Zhao et al. [11] proposed a multitasking learning architecture based on shared spans, which randomly selected representative spans from candidate spans to perform the paired extraction of aspects and opinions. Aspect-level user sentiment recognition is usually treated as a binary classification problem (positive or negative). Cambria et al. [26] integrated top-down and bottom-up learning via an ensemble of symbolic and subsymbolic AI tools to detect the type from the text. To improve sentiment classification, neutrality is key to distinguishing positive and negative sentiment polarities [27]. Moreover, Wang et al. proposed multi-level fine-scaled sentiment analysis with ambivalence handling [28]. Recently, J et al. [29] proposed using a depth model based on the attention mechanism for more fine-grained sentiment classification. In a manner similar to context analysis in conversational sentiment analysis [30], Chen et al. constructed a

single comment sentence into a question-and-answer form to obtain its internal semantic information [21]. Specifically, they transformed the ASTE task into a multi-turn machine reading comprehension task and proposed a bidirectional machine reading comprehension (MRC) framework. The unrestricted queries in this framework are used to extract aspect terms and opinion terms, and then the restricted queries are used to classify the sentiment results to generate sentiment triplets.

- **Joint extraction methods.** Since the pipeline method breaks sentiment triplet extraction into multiple subtasks for ABSA, it suffers from error propagation between different subtasks. Xu et al. proposed a new sentiment tag to jointly extract the sentiment triplets by sequence tagging [23]. For the sequence tagging method, they add position and sentiment polarity tags and propose two tagging schemes (JET^t and JET^o) based on the characteristics of aspect and opinion terms. In spite of the fact that this tagging scheme effectively reduces the error propagation caused by a pipeline, it does not utilize the relationship between aspect terms and opinion terms effectively. Considering the relationship between aspect terms and opinion terms [24,31], Zhang et al. proposed a multitask learning framework to address the limitations by highlighting the uses of joint training, decoupled aspect and sentiment prediction, and regularization among correlated tasks during learning. However, this method cannot make full use of the existing relationships, easily leading to incorrect extraction of sentiment triplets. Wu et al. proposed a tagging scheme, named the grid tagging scheme, to extract sentiment triplets in an end-to-end fashion with one unified grid tagging task [24]. However, the sequence tagging method has difficulty dealing with the complex relationship between aspect and opinion terms, such as one-to-many and many-to-one relations. Wu et al. [32] implemented a transition system with a neural design based on modeling the triplet prediction as a graph structure. As the injection of external knowledge plays an important role in ABSA [33], Chen et al. proposed a semantic and syntactic enhanced ASTE model to fully exploit the syntactic and semantic relationships between the triplet elements and jointly extract them [34].

Our proposed framework is different from the methods of ASTE mentioned above in two ways. First, we extract the sentiment triplets in one-shot based on the span method rather than the pipeline method or the sequence tagging method. Second, we introduce the syntactic-dependency information obtained by graph convolution into our model to enhance the semantic representations of spans, which is ignored in previous work on sentiment triplet extraction.

3. Methodology

3.1. Problem formulation

Given an input sentence $S_x = \{x_1, x_2, \dots, x_n\}$ with n words, let A denote the set of aspects and O denote the set of opinion terms included in S_x . Both the aspect term a_i ($a_i \in A$) and opinion term o_j ($o_j \in O$) consist of one word or multiple consecutive words in the sentence. Let s_k indicate one type of sentiment and $s_k \in S$, where $S = \{s_1, s_2, \dots, s_m\}$. Then, the ASTE task is to extract the collection of aspect sentiment triplets $T = \{(a_i, o_j, s_k) \mid (a_i, o_j, s_k) \in A \times O \times S\}$. Without loss of generality, we focus on three types of sentiment in this work: positive (POS), neutral (NEU) and negative (NEG), which are also utilized in previous works on ASTE.

3.2. Framework overview

The overall architecture of the proposed SSJE is shown in Fig. 2, and it consists of four main parts: the encoding module, the dependency graph convolutional module, the span generator and filter, and the aspect sentiment triplet extractor.

Specifically, given a review sentence, we use the bidirectional encoder representation from transformers (BERT) as our core backbone to learn the semantic representation of the context in the encoding layer. Moreover, considering the syntactic dependency between the words of the sentence, we integrate the syntactic dependency information obtained by a GCN over the syntactic dependency tree of the sentence into the representation to enhance the semantic features. Then, we enumerate all possible spans of different lengths through the span generator and filter out the invalid spans that do not contain aspect terms and opinion terms to reduce the number of candidate spans. Finally, the remaining representative spans are combined with their local context to complete ASTE by implementing a binary classification with a sentiment triplet extractor.

3.3. Encoding

The proposed SSJE uses BERT as the core backbone to learn the semantic context representation. Given an input sentence $S_x = \{x_1, x_2, \dots, x_n\}$ with n words, a special classification token $[cls]$ is appended at the beginning of the input sentence, and the separator $[sep]$ is appended at the end of the sentence. Then, the extended word sequence " $x_{cls}, x_1, x_2, \dots, x_n, x_{sep}$ " is input into the BERT to generate the word embedding sequence, and the output representation is given as follows:

$$h_i^{l-1} = BERT(x_i) \quad (1)$$

where h_i^{l-1} indicates the output of the hidden unit at the $(l-1)$ -th layer for the i th token in the BERT model. $BERT(*)$ denotes the fine-tuned BERT model, which consists of 12 hidden layers, and the features with contextual semantics are obtained through the last layer of the hidden layer.

3.4. Integrating the syntactic dependency information

There is a strong interaction between different words in one sentence. This interaction provides a crucial clue for identifying the relationship between aspect terms and opinion terms. Nevertheless, obtaining and representing the complex interactions among various words pose a challenge [35]. To address this issue, reducing the interference between different words is a critical step. In recent years, for the AOPe task, researchers have mainly solved this problem based on the attention mechanism [36], which may fail because of syntactic complexity. As shown in the example sentence in Fig. 3, the opinion term *delicious* is used to modify the aspect term *noodles*, and the opinion term *terrible* is used to modify the aspect term *vegetables*. However, because *terrible* is closer to *noodles* than *delicious*, the attention mechanism allocates more attention to *terrible* for *noodles* than *delicious*, resulting in capturing an incorrect relationship between the aspect term and opinion term, and further leading to an incorrect sentiment prediction. As shown in Fig. 3, it is evident that the syntactic dependency tree of a sentence can provide dependency information for the extraction of aspect and opinion terms. Specifically, the relationship between the aspect term *noodles* and the opinion term *delicious* is a noun subject dependency (nsubj), which indicates that there is a modified attribute or entity. Similarly, there is a conjunct dependency between the two aspect terms *noodles* and *vegetables* in this sentence, indicating that they have the same attributes or characteristics.

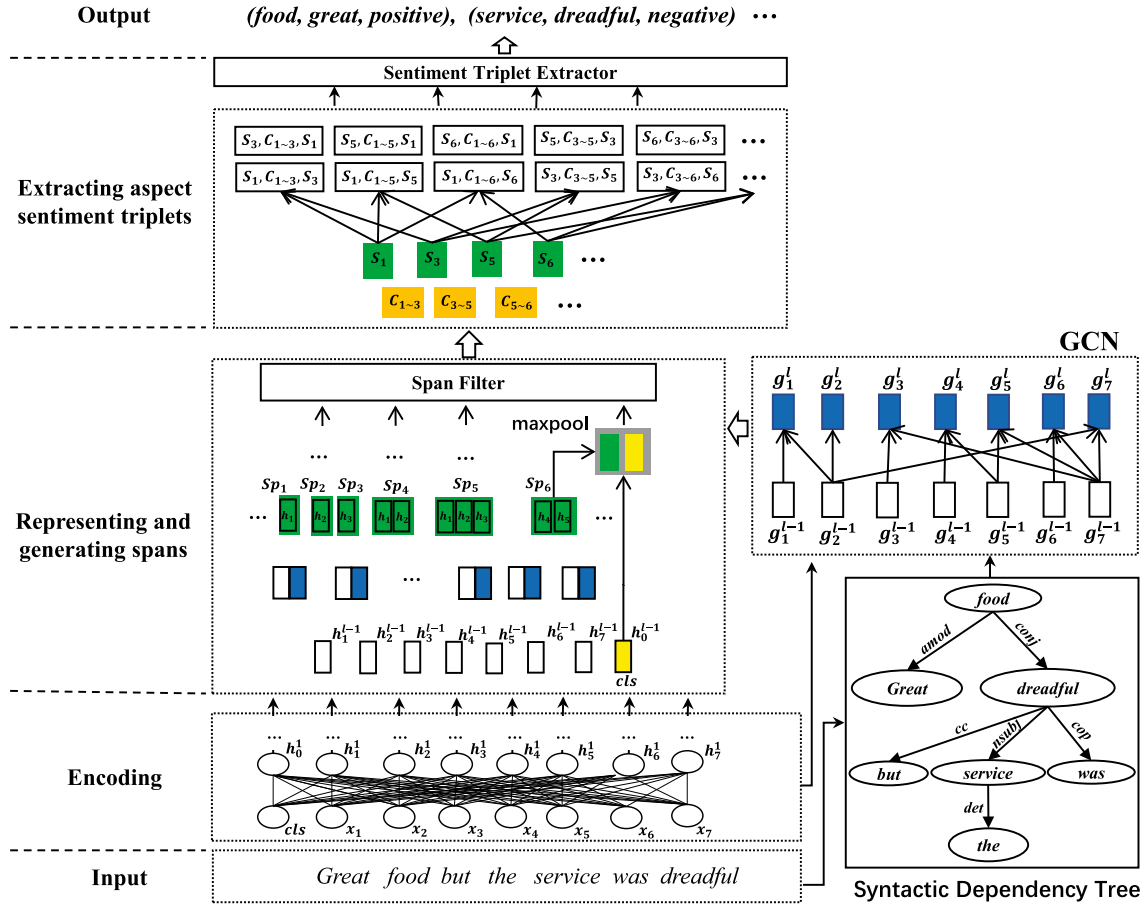


Fig. 2. The overall architecture of SSJE. The green squares represent the span features obtained by the output feature of the GCN and encoder through the max pooled layer; the orange squares represent the local context between two spans, and the yellow squares represent the global context.

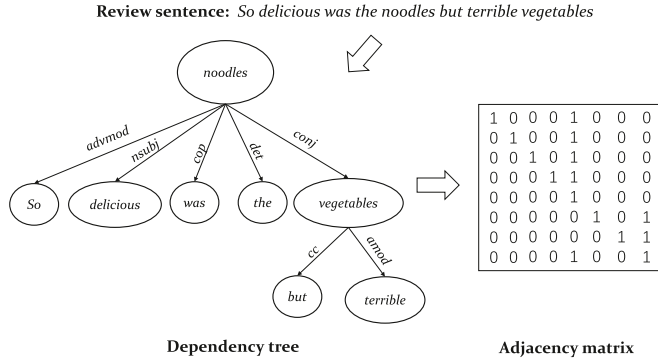


Fig. 3. Example of a syntactic dependency tree. The direction of the arrow indicates the dependency between words.

To utilize the syntactic dependency information to enhance the semantic representation of spans, we use the Stanford-NLP tool¹ to generate the syntactic dependency tree of the sentence. Then, a directional graph $G = \{V, E\}$ is used to represent words and their dependency relations, where V is the set of N nodes and E is the set of edges. Each node represents a word of the sentence, and an edge from node i to node j means there is a dependency relationship between the two corresponding words. Graph G can be expressed as an adjacency matrix A of dimensions $N \times N$, as shown in Fig. 3. We set the element a_{ij} of the adjacency matrix

as follows:

$$a_{ij} = \begin{cases} 1, & i = j \text{ or node } i \text{ is connected to node } j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Then, a GCN is implemented based on the BERT output and the adjacency matrix. Specifically, for node i at the l th ($l \geq 2$) layer, its hidden representation is obtained by the following equation:

$$g_i^l = \text{RELU} \left(\sum_{j=1}^N a_{ij} W^l g_j^{l-1} + b^l \right) \quad (3)$$

where W^l is the trainable weight matrix for the l th layer and b^l is the offset value of the current weight. When $l = 1$, g_j^0 is word j 's embedding generated by BERT.

3.5. Generating and filtering the candidate spans

Two essential issues should be given special attention for identifying the opinion entities included in a sentence:

1. How can the problem of entity overlap be addressed effectively? It is common for one aspect term to correspond to multiple opinions or one opinion term to correspond to multiple aspect terms in real-world scenarios.
2. How can the number of the generated spans be effectively reduced?

For the first issue, we enumerate all possible spans with a length constraint, and they are shared as candidate aspect terms and opinion terms simultaneously, where a span is a single word

¹ <https://stanfordnlp.github.io/CoreNLP/>

or multiple consecutive words with a starting index *start* and an ending index *end* in the sentence. One span can be repeatedly selected as a candidate aspect term or opinion term to be a potential opinion expression with another span in the form of aspect-opinion pairs. In this way, we can solve the problem of multiple-word aspect terms and opinion terms. Moreover, we can also effectively deal with one-to-many and many-to-one relations between aspect terms and opinion terms.

Specifically, we concatenate the representation h_i^{l-1} generated by BERT with the representation g_i^l (the blue block in Fig. 2) generated by the GCN to construct the feature h_i , including the context semantics and the dependency information. Then, a span generator is used to enumerate all possible spans $SP = \{Sp_1, Sp_2, \dots, Sp_n\}$ based on these features. For the span $Sp_i = \{h_{start}, h_{start+1}, \dots, h_{end}\}$, h_{start} represents the feature of the start position, and h_{end} represents the feature of the end position. It is worth noting that we set the hyperparameter L_s to limit the span's length with the constraint $0 \leq end-start \leq L_s-1$.

For spans consisting of multiple words, we fuse the feature representations of words with a max pooling layer. For example, for the span width $k+1$, the corresponding trainable width embedding vector E_{k+1} is looked up through a special embedding matrix, and the embedding matrix is trained by back-propagation. Then, a span can be expressed in the following form:

$$Sp_i = f(h_{start}, h_{start+1}, \dots, h_{end}) : E_{k+1} \quad (4)$$

where f is the max pooling function, and its result is connected with E_{k+1} .

For the second issue, since the span-based method enumerates all possible spans with different lengths, it will produce a large number of samples and lead to a huge search space. However, there are many invalid spans, such as those that do not include aspect terms or opinion terms, which will be filtered by a span filter. The span filter is a binary classifier that takes the input Sp_i together with token $[cls]$'s hidden state acting as the context of the whole sentence and outputs *yes* or *no* for each input. The label *yes* means that the corresponding span is kept; otherwise, the span is abandoned. The classification is implemented by a linear layer:

$$L_{Sp} = \text{Linear}(W^{Sp}(Sp_i : C_{cls}) + b^{Sp}) \quad (5)$$

where W^{Sp} and b^{Sp} are the trainable weight matrix and bias, respectively. Finally, the output results are normalized with a *softmax* function:

$$P(y_i | Sp_i) = \text{Softmax}(L_{Sp}) \quad (6)$$

As a result, the span filter's loss function can be formulated based on the cross-entropy error between the predicted distribution $P(y_i | Sp_i)$ and gold distribution $P(y_i^T | Sp_i)$:

$$Loss_{Sp} = - \sum_{i=1}^k P(y_i^T | Sp_i) \log(P(y_i | Sp_i)) \quad (7)$$

3.6. Extracting the aspect sentiment triplets

The refined spans serve as both candidate aspect terms and candidate opinion terms. Any combination of these spans is regarded as a potential opinion expression consisting of a pair of aspect terms and opinion terms. Notably, instead of extracting aspect-opinion pairs, we extract the aspect sentiment triplets of the aspect term, opinion term and corresponding sentiment by implementing a multiclassification task. Specifically, we classify a candidate opinion expression (Sp_i, Sp_j) as $s_{i,j}^r$, where Sp_i and Sp_j ($Sp_i, Sp_j \in SP$) are the candidate aspect term and opinion term, respectively, and $s_{i,j}^r$ ($s_{i,j}^r \in S$) indicates the sentiment.

Considering the importance of the context semantics, we introduce the context representation to the candidate sentiment triplets. The traditional BERT-based methods append the global context token $[cls]$ as the representation of the whole sentence. However, for aspect-level sentiment analysis, a sentence often contains multiple aspects or opinions. Thus, the global context is not applicable in identifying the sentiment of a pair of candidate spans. To address this, we only use the local context semantics $(C_{start}^1, \dots, C_{end}^2)$, rather than the global context, where C_{start}^1 and C_{end}^2 are representations of the first word and the last word between the two candidate spans in the sentence. A max pooling layer is used to integrate these contexts for an aspect-opinion pair:

$$C_{i,j}^k = f(C_i^1, \dots, C_j^k) \quad (8)$$

Since there can be no overlap between the aspect term and opinion term in real review sentences, we use this heuristic to filter impossible combinations to reduce the search space.

Next, we concatenate the candidate span pair with the local context as the representation of an input sample:

$$I_{rep} = [Sp_i : C_{i,j}^k : Sp_j] \quad (9)$$

To determine the sentiment type of a triplet, the sentiment triplet extractor employs a linear layer to classify its sentiment type:

$$P(\hat{s}_{i,j}^r | (Sp_i, Sp_j)) = \sigma(\text{Linear}(W^{rep}I_{rep} + b^{rep})) \quad (10)$$

where σ is the sigmoid function and W^{rep} and b^{rep} are the trainable weight matrix and bias, respectively.

There is a greater likelihood of sentiment expression between pairs of candidate spans when the $P(\hat{s}_{i,j}^r | (Sp_i, Sp_j))$ value is higher. Thus, for a sentiment triplet $(Sp_i, Sp_j, \hat{s}_{i,j}^r)$, if its score $P(\hat{s}_{i,j}^r | (Sp_i, Sp_j))$ is larger than a predefined threshold ξ , this triplet will be considered an aspect sentiment triplet, where Sp_i , Sp_j and $\hat{s}_{i,j}^r$ are the aspect term, opinion term and corresponding sentiment, respectively.

Finally, the loss function for ASTE can be formulated by the cross-entropy error between the predicted distribution $P(\hat{s}_{i,j}^r | (Sp_i, Sp_j))$ and gold distribution $P(s_{i,j}^T | (Sp_i, Sp_j))$:

$$Loss_{sc} = - \sum_{i=1}^k \sum_{j=1}^k P(\hat{s}_{i,j}^T | (Sp_i, Sp_j)) \log(P(\hat{s}_{i,j} | (Sp_i, Sp_j))) \quad (11)$$

3.7. Training procedure

In the training process of the proposed SSJE model, the loss function of joint training is defined for the span filter and the aspect sentiment triplet extractor:

$$Loss_{joint} = Loss_{Sp} + Loss_{sc} \quad (12)$$

where $Loss_{Sp}$ is the cross-entropy loss function of the span filter and $Loss_{sc}$ is the cross-entropy loss function of the aspect sentiment triplet extractor. During the training process, the following two points are worth noting:

1. In contrast to the pipeline method, in the span filter, we do not extract aspect terms or opinion terms separately but extract the candidate spans shared as aspect terms and opinion terms. By using the span filter to distinguish whether the span contains aspect terms or opinion terms, we can effectively use the relationship between aspect terms and opinion terms through the local context in the stage of sentiment triplet extraction to effectively prevent error propagation between different subtasks.

2. We perform different preprocessing to generate negative samples for the span generator and sentiment triplet extractor. When the span filter is trained, all spans labeled as aspect terms and opinion terms in the sentence are treated as positive samples, while the negative samples N_{sp} are randomly selected from the spans Sp^n that do not contain aspect terms and opinion terms:

$$N_{sp} = \text{random}(Sp^n), \quad (13)$$

$$Sp^n \notin Sp^A \wedge Sp^n \notin Sp^O$$

where Sp^A indicates the set of spans containing aspect terms and Sp^O indicates the set of spans containing opinion terms. When training the triplet extractor, we take the correctly labeled sentiment triplets as the positive samples, while the negative samples come from the entity pairs that do not have sentiments assigned. In the example sentence in Fig. 2, we choose (*food*, *great*, *POS*) and (*service*, *dreadful*, *NEG*) as positive samples and choose (*food*, *dreadful*, ***), (*food*, *service*, ***) and (*great*, *dreadful*, ***) as negative samples, where *** indicates an arbitrary type of sentiment. Compared to those of random sampling approach, the negative samples obtained by our method are more representative, and the experimental results verify that this negative sampling method greatly improves the robustness of the model.

4. Experiments

4.1. Datasets and settings

Datasets. To verify the effectiveness of our proposed SSJE framework, we conduct experiments on two benchmark datasets: the ASTE-Data-V2 dataset² and the TOWE dataset,³ which were built based on the SemEval Challenges [37–39]. In the ASTE-Data-V2 dataset, aspect sentiment triples are labeled in four subdatasets (14lap_t, 14res_t, 15res_t and 16res_t) [23]. *Res* and *lap* indicate datasets from the restaurant domain and laptop domain, respectively. Compared to the previous version ASTE-Data-V1 [20], ASTE-Data-V2 contains more complex opinion expressions. For example, an aspect term/opinion term is associated with multiple opinion terms/aspect terms, which is very common in real-world scenarios. The statistics of the ASTE-Data-V2 dataset are listed in Table 1, where #0, #+ and #- denote the numbers of aspect sentiment triplets with the neutral, positive and negative sentiments, respectively; #S and #T denote the number of sentences and aspect sentiment triplets, respectively.

Although the proposed SSJE is designed to extract aspect sentiment triplets, it is also competent for the AOPe task, i.e., extracting aspect-opinion pairs. To verify this effectiveness, we carry out experiments on the TOWE dataset, in which all aspect-opinion pairs in the form (*aspect term*, *opinion term*) are labeled in four subdatasets (14lap_p, 14res_p, 15res_p and 16res_p) [7]. The detailed statistics of the TOWE dataset are shown in Table 2, where #A, #O and #P denote the numbers of aspect terms, opinion terms and aspect-opinion pairs, and #S denotes the number of sentences.

As mentioned in Section 3.5, a classifier is used to filter the invalid spans in the proposed SSJE. To train this span filter, we construct a temporary dataset based on the above datasets. Because the aspect terms and opinion terms are labeled in the ASTE-Data-V2 dataset and the TOWE dataset, we can easily identify whether a sample contains an aspect term or an opinion term.

Table 1

Statistics of the ASTE-Data-V2 dataset for ASTE tasks.

Subdatasets		#0	#+	##-	#S	#T
14lap_t	Train	126	817	517	906	1460
	Dev	36	169	141	219	346
	Test	63	364	116	328	543
14res_t	Train	166	1692	480	1266	2338
	Dev	54	404	119	310	577
	Test	66	773	155	492	994
15res_t	Train	25	783	205	605	1013
	Dev	11	185	53	148	249
	Test	25	317	143	322	485
16res_t	Train	50	1015	329	857	1394
	Dev	11	252	76	210	339
	Test	29	407	78	326	514

Table 2

Statistics of the TOWE dataset for AOPe tasks.

Subdatasets		#S	#A	#O	#P
14lap_p	Train	1151	1626	1625	1871
	Test	343	481	498	565
14res_p	Train	1625	2639	2722	3062
	Test	500	864	888	1030
15res_p	Train	754	1076	1192	1277
	Test	325	436	469	493
16res_p	Train	1079	1512	1661	1770
	Test	328	456	485	524

Settings. In the encoding layer, we use BERT⁴ as our backbone for fine-tuning during the training process; it contains 12 attention mechanism heads, 12 hidden layers and 768 hidden units. To select an optimal combination of parameters L_s , ξ and N_{sp} , grid search is utilized, and the selection criterion is the extraction effectiveness of the validation set when we tune the proposed SSJE. In the model training phase, we use the AdamW algorithm [40] for optimization; the weight decay value is set to 0.01 and the warmup rate is 0.1. The training learning rate of the model is set to 5e-5. After embedding, the dropout is set to 0.1 to prevent overfitting in the proposed framework. The experimental workstation houses two 2.40 GHz Xeon Silver 4210R CPUs, 32 GB of RAM and an NVIDIA Tesla V100 graphics card containing 16 GB of device memory. The operating system is Linux 4.15.0-107-generic in our experiments. For the convenience of reproduction, we make our codes⁵ publicly available.

Evaluation metrics. We use the precision (P), recall (R) and F_1 scores as the evaluation indicators to measure the effectiveness of ASTE in our experiments:

$$P = \frac{|TE|}{|E|} \quad (14)$$

$$R = \frac{|TE|}{|Trip|} \quad (15)$$

$$F_1 = \frac{2 \times P \times R}{P + R} \quad (16)$$

where TE is the set of triplets correctly extracted by the model, E is the set of triplets extracted by the model, and $Trip$ is the set of triplets included in the dataset. For AOPe, we use similar evaluation indicators.

Baselines. To demonstrate the effectiveness of our model, we conducted comparative experiments with the baselines below for

² <https://github.com/xuuluuuu/SemEval-Triplet-data/tree/master/ASTE-Data-V2-EMNLP2020>

³ <https://github.com/NJUNLP/TOWE>

⁴ <https://github.com/google-research/bert>

⁵ <https://github.com/MKMaS-GUET/SSJE>

ASTE, which can be summarized in two groups: pipeline methods and joint extraction methods.

1. Pipeline methods

- **CMLA+** [16] (Wang et al. 2017): CMLA is a serialization labeling method. CMLA uses the attention mechanism to extract the relationship between aspects and opinions. CMLA+ adds a multilayer perceptron (MLP) to CMLA to further determine whether the triplets are correct in the matching phase.
- **RINANTE+** [18] (Dai et al. 2019): RINANTE captures word dependencies in sentences based on LSTM-CRF and some fusion rules. The method of extracting the sentiment triplets is the same as that of CMLA+.
- **Li-unified-R+** [25] (Li et al. 2019): Li-unified-R is used to assist target boundary detection and complete aspect-based sentiment prediction by stacking two LSTMs, which is also the sequence labeling method. Finally, the same method as in CMLA+ is used to obtain triplets.
- **Peng et al.** [20] (2020) proposed a two-step framework to extract aspect sentiment triplets, in which aspect words and the corresponding sentiment polarities are extracted in the first stage and each aspect with its sentiment type is paired with the corresponding opinion expression in the second stage.
- **BMRC** [21] (Chen et al. 2021) transforms the ASTE task into a multiturn machine reading comprehension task, and builds associations among different ABSA subtasks with three types of queries, including non-restrictive extraction queries, restrictive extraction queries and sentiment classification queries, to obtain aspect sentiment triplets.

2. Joint extraction methods

- **OTE-MTL** [31] (Zhang et al. 2020) uses a multitask learning framework for jointly extracting aspect sentiment triplets, which jointly extracts aspect terms and opinion terms, simultaneously parses the sentiment dependencies between them with a biaffine scorer and generates sentiment triplets by label serialization.
- **JET** [23] (Xu et al. 2020) uses a new location-aware labeling scheme based on the *BIOES* tagging scheme, which adds location and sentiment information to complete sentiment triplet extraction. This method has two tagging schemes: target-based joint extraction triplets (JET^t) and opinion-based joint extraction triplets (JET^o).
- **GTS** [24] (Wu et al. 2020) uses a uniform grid tagging mechanism for each aspect-opinion pair to be assigned to a specific representation of the relationship.

4.2. Results and discussion

As mentioned previously, although the proposed SSJE is designed to extract the aspect sentiment triplets from review sentences, it is also competent for performing AOPE. We verify SSJE's effectiveness on the ASTE task and AOPE task in the following experiments.

4.2.1. Effect of aspect sentiment triplet extraction

We compare the proposed SSJE with the pipeline methods and the joint extraction methods for extracting aspect sentiment triplets on the ASTE-Data-V2 dataset. The experimental

results are shown in Table 3. We find that SSJE achieves the best extraction effectiveness on all four subdatasets. Compared to other pipeline baselines, BMRC performs better; it transforms the sentiment triplet extraction task into a machine reading comprehension task. Although this method can solve the one-to-many and many-to-one problems that are difficult to solve by the sequence tagging method, it needs to extract the correct aspects and opinion terms first. Then, BMRC suffers from error propagation in different subtasks and results in relatively poor extraction performance. The average P , R and F_1 scores of SSJE are 3.32%, 3.97% and 3.58% higher than those of the best pipelined baseline (i.e., BMRC), respectively. We think the main reason is that SSJE extracts the aspect terms, opinion terms and corresponding sentiment simultaneously, which can effectively prevent the error propagation.

Nevertheless, GTS achieves better effectiveness than other joint extraction baselines because GTS assigns a specific relationship to each aspect-opinion pair, which makes use of the partial relationship between aspect terms and opinion terms. Since GTS needs to semantically annotate each word, it cannot deal with the one-to-many and many-to-one relationships between aspect terms and opinion terms. However, the proposed SSJE effectively solves this entity overlap problem by sharing the candidate spans between aspect terms and opinion terms at the same time. The average P , R and F_1 scores of SSJE are 9.38%, 11.53% and 10.45% higher than those of the best joint extraction baseline (i.e., GTS), respectively.

4.2.2. Effect of aspect-opinion pair extraction

To verify that the proposed SSJE is also competent for AOPE task, we compare it with two state-of-the-art baselines: SpanMlt (Zhao et al. 2020 [11]) and GTS (Wu et al. 2020 [24]) for extracting aspect-opinion pairs on the TOWE dataset. The experimental results are shown in Table 4.

Compared to GTS, the average P , R and F_1 scores are 7.51%, 4.69% and 6.13% higher than those of GTS, respectively. The main reason is that the span-based method used by SSJE can deal with the entity overlap problem, which is difficult for sequence tagging-based methods such as GTS. In addition, we compare SSJE with another span-based model spanMlt. As shown in Table 4, SSJE achieves better effectiveness than spanMlt. Specifically, the average P , R and F_1 scores of SSJE have improved by 3.68%, 4.74% and 4.31%, respectively, compared to spanMlt. We think the main reason is that SSJE incorporates syntactic dependency information to enhance the spans' semantic representation, which is helpful for improving extraction effectiveness.

4.3. Framework analysis

4.3.1. Ablation study

In the proposed SSJE, the dependency information generated by a GCN based on the sentence's syntactic dependency tree is incorporated to enhance the span representations. To verify the effectiveness of integrating this information, we conduct an ablation study on the ASTE task and the AOPE task. The experimental results are shown in Table 5, where SSJE w/o GCN represents the SSJE model without integrating the dependency information and the symbol $\downarrow$ ($\uparrow$) indicates a decrease (improvement) over the effect of SSJE.

When the dependency information is not integrated into the model, the extraction effect of SSJE declines for both the ASTE task and AOPE task. In the ASTE task, the average F_1 score of SSJE on four subdatasets of the ASTE-Data-V2 dataset decreases by 1.56 when the GCN module is removed from SSJE. In particular, on the 15res_t subdataset, the F_1 score of SSJE decreases from 65.05% to 62.55%. In the AOPE task, when we remove the GCN module

Table 3

The experimental results (%) on the ASTE task. The best scores are highlighted in bold.

Methods		ASTE-Data-V2 dataset											
		14lap_t			14res_t			15res_t			16res_t		
		P	R	F ₁	P	R	F ₁	P	R	F ₁	P	R	F ₁
Pipeline	CMLA+	30.09	36.92	33.16	39.18	47.13	42.79	34.56	39.84	37.01	41.34	42.1	41.72
	RINANTE+	21.71	18.66	20.07	31.42	39.38	34.95	29.88	30.06	29.97	25.68	22.30	23.87
	Li-unified-R+	40.56	44.28	42.34	41.04	67.35	51.00	44.72	51.39	47.82	37.33	54.51	44.31
	Peng et al.	37.38	50.38	42.87	43.24	63.66	51.46	48.07	57.51	52.32	46.94	64.24	54.21
	BMRC	64.55	53.88	58.73	70.67	69.61	70.13	63.10	60.88	61.97	67.86	69.46	68.65
Jointextraction	OTE-MTL	49.37	43.09	46.02	66.42	55.13	60.25	64.80	40.62	49.94	57.66	59.34	58.48
	JET ^r	53.53	43.28	47.86	63.44	54.12	58.41	68.20	42.89	52.66	65.28	51.95	57.86
	JET ^o	55.39	47.33	51.04	70.56	55.94	62.40	64.45	51.94	57.53	70.42	58.37	63.83
	GTS	57.12	53.42	55.21	71.76	59.09	64.81	54.71	55.05	54.88	65.89	66.27	66.08
SSJE (Ours)		67.43	54.71	60.41	73.12	71.43	72.26	63.94	66.17	65.05	70.82	72.00	71.38

Table 4

The experimental results (%) on APOE task. The best scores are highlighted in bold.

Methods		TOWE dataset											
		14lap_p			14res_p			15res_p			16res_p		
		P	R	F ₁	P	R	F ₁	P	R	F ₁	P	R	F ₁
GTS		64.85	63.30	64.07	72.59	76.77	74.62	67.19	69.53	68.34	70.89	76.64	73.65
SpanMlt		71.98	64.59	68.09	74.24	79.76	76.90	69.20	66.53	67.84	71.51	75.19	73.30
SSJE (Ours)		72.18	72.84	72.51	75.72	80.46	78.02	70.10	68.97	69.53	79.88	78.05	78.96

Table 5

Ablation study on the ASTE task and the APOE task (%).

ASTE					APOE				
ASTE-Data-V2 dataset			methods		TOWE dataset			methods	
			SSJE w/o GCN	SSJE				SSJE w/o GCN	SSJE
14lap_t	P		62.70(↓ 4.73)	67.43	14lap_p	P		70.70(↓ 1.48)	72.18
	R		56.56(↑ 1.85)	54.71		R		70.83(↓ 2.01)	72.84
	F1		59.48(↓ 0.93)	60.41		F1		70.76(↓ 1.75)	72.51
14res_t	P		73.45(↑ 0.33)	73.12	14res_p	P		73.95(↓ 1.77)	75.72
	R		69.32(↓ 2.11)	71.43		R		80.26(↓ 0.20)	80.46
	F1		71.33(↓ 0.93)	72.26		F1		76.97(↓ 1.05)	78.02
15res_t	P		61.43(↓ 2.51)	63.94	15res_p	P		65.95(↓ 4.16)	70.10
	R		63.71(↓ 2.46)	66.17		R		67.95(↓ 1.02)	68.97
	F1		62.55(↓ 2.50)	65.05		F1		66.93(↓ 2.60)	69.53
16res_t	P		69.01(↓ 1.81)	70.82	16res_p	P		76.60(↓ 3.28)	79.88
	R		70.62(↓ 1.38)	72.00		R		77.48(↓ 0.57)	78.05
	F1		69.81(↓ 1.57)	71.38		F1		77.04(↓ 1.92)	78.96

from SSJE, the extraction effectiveness of SSJE also declines on all four subdatasets. In particular, compared with SSJE, the F_1 score of SSJE w/o GCN decreases by 2.60 on the 15res_p subdataset. Overall, when the GCN module is removed from SSJE, the average F_1 score of SSJE decreases by 2.32% for the ASTE task and 2.43% for the APOE task. Therefore, integrating dependency information into the model to enhance the semantic representation of spans is beneficial for improving the effect of aspect sentiment extraction and aspect-opinion extraction.

4.3.2. Parameter sensitivity

There are three hyperparameters in the proposed SSJE: the maximum span length L_s , the confident threshold ξ for the sentiment triplet extractor and the number N_{sp} of negative samples for training the span filter and the sentiment triplet extractor. In this experiment, we investigate the effects of these three parameters on the effectiveness of ASTE. It is worth noting that although we use the method of grid parameter adjustment to automatically select the optimal combination of L_s , ξ and N_{sp} , in order to observe the effect of the parameters on the model, we fix one, adjust the other, and observe the effect.

The maximum span length (L_s). We vary L_s in the range of [1, 9] with a step of 1 while fixing the threshold ξ at 0.6 and

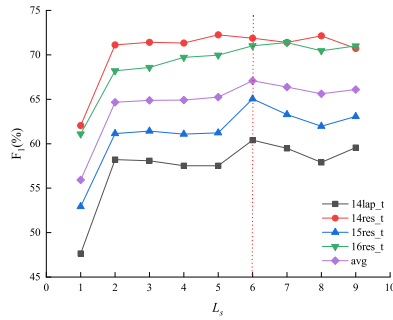
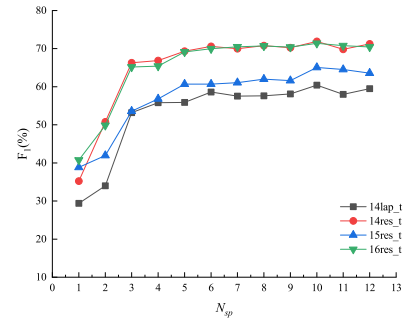
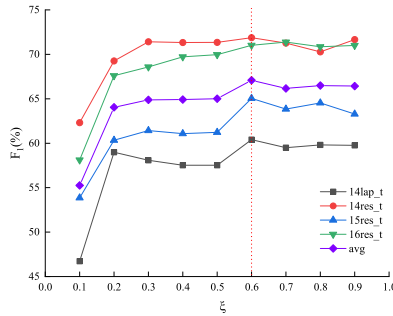
N_{sp} at 10. In Fig. 4, we observe that a span of appropriate length is necessary. The extraction performance is very poor when $L_s = 1$ because aspect terms and opinion terms only contain one word in the sentence in this case. With the gradual increase in the span's maximum length, the extraction performance of SSJE is gradually improved. However, when L_s reaches 6, the performance change of extraction tends to be flat. Considering that the training time of the model increases with increasing span length, we think that setting L_s to 6 is generally an appropriate choice.

The confidence threshold (ξ). As presented in Section 3.6, we filter the aspect sentiment triplets with low confidence by a predefined threshold ξ . In this experiment, we vary ξ in the range of [0.1, 0.9] with a step of 0.1 while fixing L_s at 6 and N_{sp} at 10. In Fig. 5, when ξ is set to 0.1, SSJE performs poorly. This is because when the threshold is too small, a large number of invalid sentiment triplets cannot be filtered out effectively. According to the performance of the average curve, when $\xi = 0.6$, the extraction performance of SSJE tends to be stable.

The negative sample number (N_{sp}). As mentioned in Section 3.7, we construct negative samples for the span filter and sentiment triplet extractor with different strategies. Since the number of negative samples is determined by the positive samples for the sentiment triplet extractor, we discuss only the effect

Table 6Case study of ASTE. Incorrect results are marked with $\times$.

Sentences	Ground truth	BMRC	GTS	SSJE (Ours)
S1. Great food but the service was dreadful!	{{(food, great, POS), (service, dreadful, NEG)}}	{{(food, great, POS), (service, dreadful, NEG)}}	{{(food, great, POS), (service, dreadful, NEG)}}	{{(food, great, POS), (service, dreadful, NEG)}}
S2. ! so nice.. stable.. fast .. now i got my ssd !	{{(ssd, fast, POS), (ssd, nice, POS), (ssd, stable, POS)}}	{{ $\times$, (stable, nice, POS) $\times$, $\times$ }}	{{(ssd, fast, POS), (ssd, nice, POS), $\times$ }}	{{(ssd, fast, POS), (ssd, nice, POS), (ssd, stable, POS)}}
S3. the built quality is really good, i was so happy and excited about this product.	{{(built quality, good, POS), (built quality, happy, POS), (built quality, excited, POS)}}	{{(built quality, good, POS), $\times$, $\times$ }}	{{(built quality, good, POS), (built quality, happy, POS), $\times$ }}	{{(built quality, good, POS), (built quality, happy, POS), (built quality, excited, POS)}}
S4. Love the enchiladas and chicken soup - and be sure to check out their specials.	{{(enchiladas, Love, POS), (chicken soup, Love, POS)}}	{{(enchiladas, Love, POS), $\times$ }}	{{(enchiladas, Love, POS), (chicken soup, Love, POS), (specials, Love, POS) $\times$ }}	{{(enchiladas, Love, POS), (chicken soup, Love, POS)}}
S5. the performance seems quite good, and built-in applications like iphoto work great with my phone and camera.	{{(iphoto, great, POS), (built-in applications, great, POS), (performance, good, POS)}}	{{ $\times$, $\times$, (performance, good, POS)}}	{{(iphoto, great, POS), (built-in applications, great, POS), (performance, good, POS), (the performance, iphoto, POS) $\times$ }}	{{(iphoto, great, POS), (built-in applications, great, POS), (performance, good, POS)}}

**Fig. 4.** Effect of the maximum span length L_s .**Fig. 6.** Effect of the negative sample number N_{sp} .**Fig. 5.** Effect of the confidence threshold ξ .

of the negative sample number N_{sp} for the span filter. Fig. 6 shows the F_1 scores for different numbers of negative samples generated from a sentence, where $\xi = 0.6$ and $L_s = 6$. When N_{sp} is set to 1, the effectiveness of sentiment triplet extraction is very poor. For example, the F_1 score is only 29.40% for the 14lap_t dataset and 35.22% for the 14res_t dataset. However, when N_{sp} increases to 10, the performance of the model continues to improve and tends to be stable. Most review sentences are relatively short. Specifically, the sentence length is not more than 50 words in our datasets. Therefore, the number of negative samples that do not contain aspect terms or opinion terms in the sentence is often relatively small. As shown in Fig. 6, we found our results to be more stable when setting the maximum N_{sp} for a sentence to 10.

4.3.3. Case study

To further demonstrate the effectiveness of the proposed SSJE, we choose some examples in the ASTE-Data-V2 dataset and show

the extracted sentiment triplets of different models in Table 6. For the convenience of writing, we only compare SSJE with the best pipeline baseline (i.e., BMRC) and the best joint extraction baseline (i.e., GTS).

In the first sentence S1, the one-to-one relationship between aspect terms and opinion terms is relatively simple. In this case, BMRC, GTS and SSJE can correctly identify the opinion expressions and their corresponding sentiments. For sentences S2 and S3, there is a one-to-many relationship between the aspect term and opinion term. BMRC fails to identify this opinion expression, especially for S2, it cannot extract any correct sentiment triplet. GTS correctly extracts two sentiment triplets from S2 and S3, but (ssd, stable, POS) is missed for S2 and (built quality, excited, POS) is missed for S3. However, the proposed SSJE accurately extracts all sentiment triplets included in S2 and S3. Therefore, SSJE can effectively address the one-to-many relationship between the aspect term and opinion term.

Moreover, sentences S4 and S5 involve a situation in which multiple aspect terms correspond to an opinion term. BMRC extracts only part of the results. The extraction results of GTS contain two errors, and it even erroneously takes the aspect term *iphoto* as an opinion term for S5. The case of the last two sentences shows that the proposed SSJE can also deal with the many-to-one relationship between the aspect term and opinion term. Furthermore, BMRC performs more poorly for the last four example sentences than GTS. The main reason is that, as a pipeline method, BMRC makes incorrect predictions in its first subtask for these relatively complex sentences, which leads to the final incorrect extraction results. Finally, we find that BMRC, GTS and SSJE can solve the problem of multiword terms, such as *built quality* in S3.

5. Conclusions

In this paper, we propose a SSJE framework to extract aspect sentiment triplets from review sentences, which provides a complete solution for ABSA. The proposed SSJE identifies the aspect term, opinion term and sentiment on the aspect simultaneously in one shot, and by this means, the error propagation in different subtasks encountered by the pipeline methods can be effectively prevented. Moreover, SSJE enumerates all possible pairs shared by aspect terms and opinion terms, which effectively deals with the situation of multiword aspect terms and opinion terms and one-to-many and many-to-one relations between aspect terms and opinion terms. Moreover, we introduce dependency information obtained by a graph neural network from the dependency tree of the sentence to enhance the semantic representations of spans. The experimental results indicate that our proposed framework achieves state-of-the-art performance, not only for ASTE tasks but also for AOPe tasks.

In future work, we can further improve our method in the following respects: first, the dependency information of words can be refined by reducing noisy information due to imperfect parsing performance and the casualness of review sentences; second, external sentiment knowledge, such as sentiment dictionaries, can be incorporated into our framework as a learning source.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the editor and anonymous reviewers for their valuable comments and feedbacks. This work was supported by the National Natural Science Foundation of China (Nos. U1811264, 62062027 and 61966009), the Guangxi Natural Science Foundations, China (Nos. 2020GXNSFAA159012 and 2018GXNSFDA281049), the Science and Technology Major Project of Guangxi Province, China (No. AA19046004), the Innovation Project of Guangxi Graduate Education, China (No. YCBZ2021069), the project of Guangxi Key Laboratory of Trusted Software, China and the project of Guilin scientific research and technology development, China (No. 2020010304).

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